Ultrametric Quantum Error Correction

Testing Discrete Geometry in the Redundancy–Fidelity Tradeoff

CWI Summer School on Quantum Algorithms & Quantum Error Correction · 24–28 August 2026 ·
Poster Session: Wed 26 Aug, 16:30–18:00 · Poster Award Ceremony: Thu 27 Aug, 08:50

Panel 1 — The Hypothesis

- **Standard view:** QEC assumes noise accumulates over a continuous topology; logical fidelity improves smoothly as redundancy is added.
- **Alternative to test:** error propagation follows a discrete, hierarchical geometry — a Bruhat–Tits tree T_p — and redundancy quantizes in units of $(p+1)$.
- **Signature to look for:** a staircase — flat plateaus followed by sudden jumps — instead of a smooth curve.

STATUS: HYPOTHESIS ONLY — NO EXPERIMENTAL RESULT CLAIMED

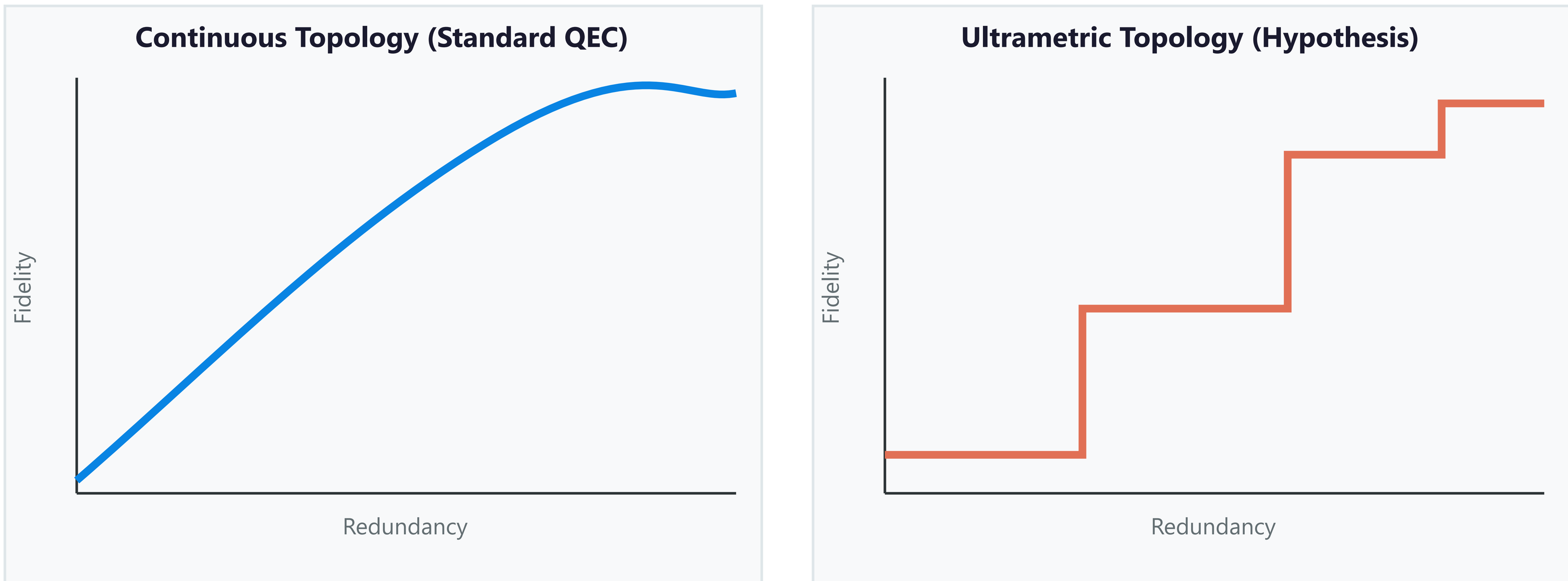
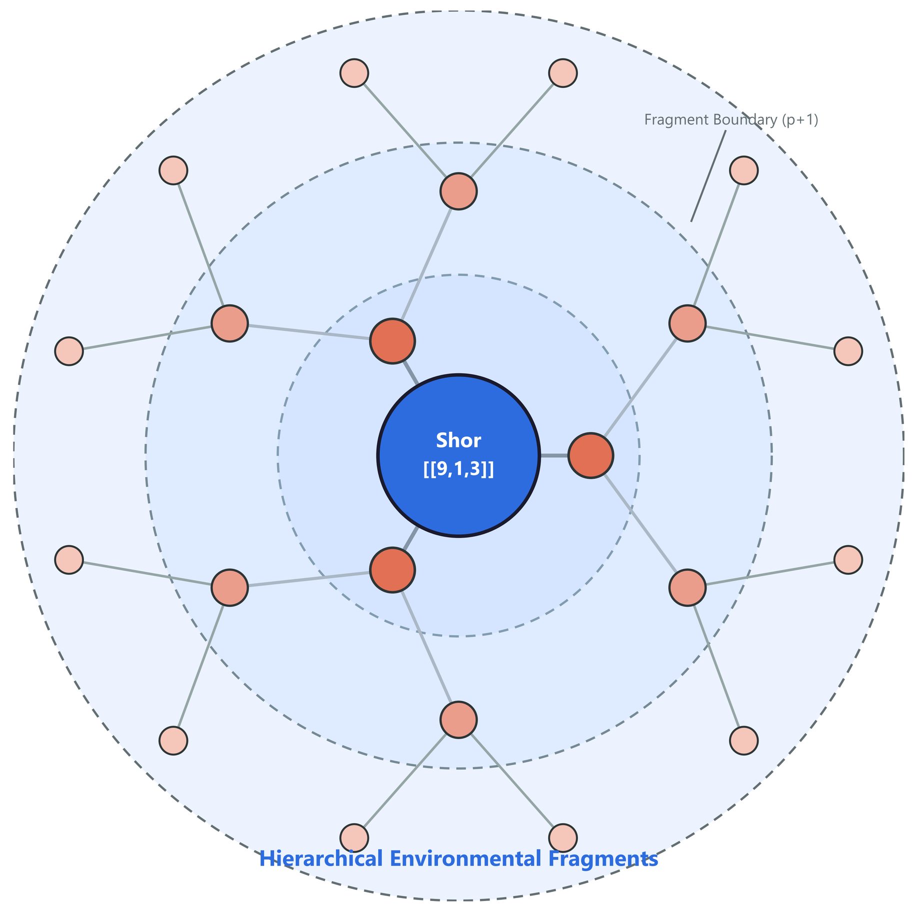


Figure 1: smooth fidelity curve (continuous topology, left) vs. staircase (ultrametric hypothesis, right). Hypothesis only; not observed.

Panel 2 — A Proposed Experiment (not yet run)



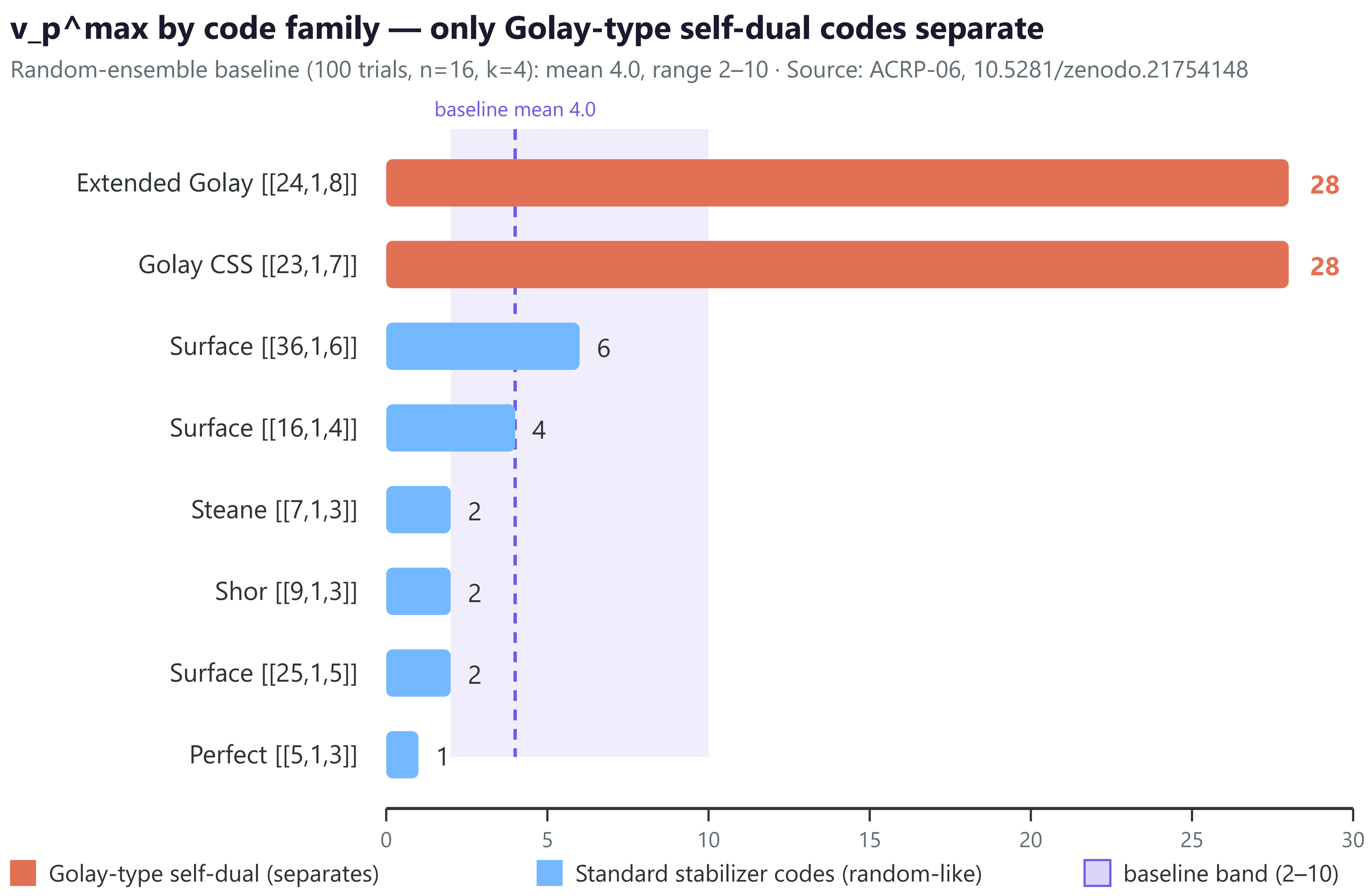
1. **Encode:** one logical qubit in the Shor [[9,1,3]] code.
2. **Interact:** expose it to a fixed, structured environment of spectator qubits.
3. **Correct:** run the full error-correction circuit; measure final logical fidelity.
4. **Scale:** group environment qubits into larger hierarchical fragments; track fidelity across fragment boundaries.

Smooth shift with fragment size = continuous model. Staircase = ultrametric model.

> **What exists today:** a provisional numerical toy model (Archimedean Shadows) suggested redundancy may quantize in discrete steps (earlier versions reported 3, 6, 12, 24) — reported provisionally — and its own most seductive signature (the "forbidden window") **failed its own falsification test**. Null reported, not hidden.

Panel 3 — The Cross-Check I Already Ran (it mostly failed — honestly)

Witness idea: 2-adic valuations as a structural fingerprint. Conjecture C7.3: the Mahler-spectrum valuation v_p^{max} separates "optimal" codes (28) from random codes (≈4).



Code family	v_p^{max}
Golay CSS [[23,1,7]]	**28**
Extended Golay [[24,1,8]]	**28**
Perfect [[5,1,3]]	1
Steane [[7,1,3]]	2
Shor [[9,1,3]]	2
Surface [[16,1,4]] / [[25,1,5]] / [[36,1,6]]	4 / 2 / 6
Random ensemble (100 trials, n=16, k=4)	mean 4.0, range 2–10

Result: the conjecture does NOT generalize. Shor, Steane, perfect and surface codes are indistinguishable from the random baseline. The 28-vs-4 gap is specific to Golay-type self-dual symmetry — not a general "optimality" witness.

> **What it teaches:** the first witness mostly failed, and the failure is informative — structure lives in an extreme-symmetry corner. **Open:** is there a refined witness (valuations of the full weight distribution $v_2(A_j)$, not just the Mahler maximum) that discriminates where v_p^{max} fails — and does any witness connect to the staircase?

Panel 4 — Value & Open Questions

A smooth curve validates continuous models; a staircase would force code architectures designed for hierarchical noise.

- What physical mechanism would produce tree-like error geometry?
- Does any valuation witness predict where the staircase should be most visible? (v_p^{max} does not.)
- How would discrete geometry alter asymptotic fault-tolerance thresholds?

I'm here to learn: has anyone in this room seen staircase-like scaling — or a reason it must be wrong?

References (Zenodo concept DOIs — verified 2026-08-16)
Ultrametric Code Spaces — 10.5281/zenodo.21824194
Archimedean Shadows (v1.11) — 10.5281/zenodo.21809888
Ultrametric Foundations (C7.3) — 10.5281/zenodo.21193003
ACRP-06 v_p^max extension — 10.5281/zenodo.21737221
Adelic Cross-Domain Program v5 — 10.5281/zenodo.21691414
"What Remains" — 10.5281/zenodo.21922812

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